w11-1、定义径向动量算符：

$$p_r = \frac{1}{2}\left(\boldsymbol{e}_r \cdot \boldsymbol{p} + \boldsymbol{p} \cdot \boldsymbol{e}_r\right)$$

其中 $\boldsymbol{e}_r = \boldsymbol{r}/r$。证明: (1) p_r 是厄米的; (2) $p_r = -i\hbar\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)$; (3) $[r, p_r] = i\hbar$; (4) $p_r^2 = -\hbar^2 \frac{1}{r}\frac{\partial^2}{\partial r^2} r$。

答案:

(1) 由于 $(ABC)^\dagger = C^\dagger B^\dagger A^\dagger$, 故

$$\begin{aligned}
p_r^\dagger &= \frac{1}{2}\left(\frac{1}{r}\boldsymbol{r}\cdot\boldsymbol{p} + \boldsymbol{p}\cdot\boldsymbol{r}\frac{1}{r}\right)^\dagger \\
&= \frac{1}{2}\left(\boldsymbol{p}^\dagger\cdot\boldsymbol{r}^\dagger\left(\frac{1}{r}\right)^\dagger + \left(\frac{1}{r}\right)^\dagger\boldsymbol{r}^\dagger\boldsymbol{p}^\dagger\right) \\
&= \frac{1}{2}\left(\boldsymbol{p}\cdot\boldsymbol{r}\frac{1}{r} + \frac{1}{r}\boldsymbol{r}\cdot\boldsymbol{p}\right) \\
&= p_r
\end{aligned}$$

(2)

$$\begin{aligned}
p_r\psi &= \frac{1}{2}\left(\frac{\boldsymbol{r}}{r}\cdot\boldsymbol{p} + \boldsymbol{p}\cdot\frac{\boldsymbol{r}}{r}\right)\psi \\
&= \frac{1}{2}\left(\frac{\boldsymbol{r}}{r}\cdot\boldsymbol{p}\psi + \boldsymbol{p}\cdot\left(\frac{\boldsymbol{r}}{r}\psi\right)\right) \\
&= -\frac{i\hbar}{2}\left(2\frac{\boldsymbol{r}}{r}\cdot\nabla\psi + \psi\nabla\cdot\frac{\boldsymbol{r}}{r}\right) \\
&= -\frac{i\hbar}{2}\left\{2\frac{\boldsymbol{r}}{r}\cdot\nabla\psi + \psi\left[\left(\nabla\frac{1}{r}\right)\cdot\boldsymbol{r} + \frac{1}{r}\nabla\cdot\boldsymbol{r}\right]\right\} \\
&= -\frac{i\hbar}{2}\left\{2\frac{\boldsymbol{r}}{r}\cdot\nabla\psi + \psi\left[\frac{-\boldsymbol{r}}{r^3}\cdot\boldsymbol{r} + \frac{3}{r}\right]\right\} \\
&= -i\hbar\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\psi
\end{aligned}$$

(3)

$$\begin{aligned}
[r, p_r]\,\psi &= -i\hbar\left[r, \frac{\partial}{\partial r} + \frac{1}{r}\right]\psi \\
&= -i\hbar\left[r\left(\frac{\partial}{\partial r} + \frac{1}{r}\right) - \left(\frac{\partial}{\partial r} + \frac{1}{r}\right)r\right]\psi \\
&= -i\hbar\left[r\frac{\partial\psi}{\partial r} - \frac{\partial}{\partial r}(r\psi)\right] \\
&= i\hbar\psi
\end{aligned}$$

(4)

$$
\begin{aligned}
p_r^2\psi &= -\hbar^2\left[\left(\frac{\partial}{\partial r}+\frac{1}{r}\right)\left(\frac{\partial}{\partial r}+\frac{1}{r}\right)\right]\psi \\
&= -\hbar^2\left[\frac{\partial^2}{\partial r^2}+\frac{\partial}{\partial r}\frac{1}{r}+\frac{1}{r}\frac{\partial}{\partial r}+\frac{1}{r^2}\right]\psi \\
&= -\hbar^2\left[\frac{\partial^2\psi}{\partial r^2}+\left(-\frac{1}{r^2}\psi\right)+\frac{2}{r}\frac{\partial\psi}{\partial r}+\frac{\psi}{r^2}\right] \\
&= -\hbar^2\left[\frac{\partial^2}{\partial r^2}+\frac{2}{r}\frac{\partial}{\partial r}\right]\psi \\
-\hbar^2\frac{1}{r}\frac{\partial^2}{\partial r^2}(r\psi) &= -\hbar^2\frac{1}{r}\left(2\frac{\partial\psi}{\partial r}+r\frac{\partial^2\psi}{\partial r^2}\right) \\
&= -\hbar^2\left[\frac{\partial^2}{\partial r^2}+\frac{2}{r}\frac{\partial}{\partial r}\right]\psi
\end{aligned}
$$

w11-2、若考虑二维径向波函数在 $r=0$ 附近的渐近解 $R(r)\sim r^a$，束缚态平方可积要求 a 的取值满足什么条件?

答案: 平方可积要求，极坐标系下径向的积分:

$$\int_0^{r_0} r|R(r)|^2dr=\int_0^{r_0} r^{2a+1}dr=r^{2a+2}\Big|_0^{r_0}=r_0^{2a+2}+0^{2a+2}$$

为有限值，即 $2a+2>0,\ a>-1$。

w11-3、质量为 μ 的粒子在如下势场中运动:

$$V(r)=\begin{cases}0, & R_1<r<R_2\\ \infty, & r\le R_1, r\ge R_2\end{cases}$$

求其 $l=0$ 的能级和相应的本征波函数。

答案: 球壳内 $l=0$ 的径向方程:

$$\frac{d^2\chi(r)}{dr^2}+k^2\chi(r)=0,\quad \left(k=\sqrt{2mE/\hbar^2}, E>0\right)$$

形式解为:

$$\chi(r)=A\sin(kr+\delta)$$

根据边界边界条件 $\chi(R_1)=\chi(R_2)=0$ 得:

$$\begin{cases} A\sin(kR_1+\delta)=0 \\ A\sin(kR_2+\delta)=0 \end{cases}$$

解得

$$k=\frac{n\pi}{R_2-R_1},\ \delta=-\frac{R_1}{R_2-R_1}n\pi$$

于是

$$E_n=\frac{n^2\pi^2\hbar^2}{2m(R_2-R_1)^2}\quad n=1,2,3,\cdots$$
$$\chi_n(r)=\sqrt{\frac{2}{R_2-R_1}}\sin\left(\frac{n\pi}{R_2-R_1}(r-R_1)\right)$$

系数 A 由归一化条件

$$\int_{R_1}^{R_2}\chi_n^2(r)dr=1$$

可得。完整的本征波函数为 $r\chi_{n0}(r)Y_{00}(\theta,\varphi)$, 即:

$$\psi_{n00}=\begin{cases} \frac{1}{r\sqrt{2\pi(R_2-R_1)}}\sin\frac{n\pi(r-R_1)}{R_2-R_1}, & R_1<r<R_2 \\ 0,\ \text{others} \end{cases}$$

w11-4、三维各向同性谐振子处于本征态 $|Nlm\rangle$，求该状态下的 $\langle r^2\rangle$，$\langle p^2\rangle$ 和 $\langle r^{-2}\rangle$ 。提示：利用费曼海尔曼定理和位力定理。

答案：三维各向同性谐振子的径向哈密顿算符:

$$H=\frac{p_r^2}{2\mu}+\frac{l(l+1)\hbar^2}{2\mu r^2}+\frac{1}{2}\mu\omega^2r^2$$

根据费曼海尔曼定理，对 ω 求导, 有

$$\frac{\partial H}{\partial\omega}=\mu\omega r^2,\frac{\partial E_n}{\partial\omega}=(N+3/2)\hbar$$

因此:

$$\langle r^2\rangle_{Nlm}=(N+3/2)\frac{\hbar}{\mu\omega}$$

或根据位力定理，对谐振子势有 $\langle V\rangle_{Nlm}=E_N/2$, 即

$$\left\langle \frac{1}{2}\mu\omega^2 r^2 \right\rangle_{Nlm}=\frac{1}{2}(N+3/2)\hbar\omega$$

整理得相同结果。

根据费曼海尔曼定理，对 l 求导，有

$$\frac{\partial H}{\partial l}=(l+1/2)\frac{\hbar^2}{\mu}\frac{1}{r^2},\frac{\partial E_N}{\partial l}=\frac{\partial E_N}{\partial N}=\hbar\omega$$

因此:

$$\left\langle r^{-2}\right\rangle_{Nlm}=\frac{1}{l+1/2}\left(\frac{\mu\omega}{\hbar}\right)$$

由于 $\langle V\rangle_{Nlm}=E_N/2$, 因此 $\langle T\rangle_{Nlm}=E_N/2$。于是有:

$$\left\langle p^2\right\rangle_{Nlm}=\left(N+\frac{3}{2}\right)\mu\hbar\omega$$

w12-5、考虑三维各向同性的谐振子，

(1) 分别写出 $N=1$ 时，(H_x,H_y,H_z) 表象和 (H,l^2,l_z) 表象的三个基矢在球坐标系下的表达式。

(2)验证变换矩阵的矩阵元 $\langle n_r=0,l=1,m=1|n_x=0,n_y=1,n_z=0\rangle=-i/\sqrt{2}$。若粒子处于状态 $|\alpha\rangle=1/2|000\rangle-i/3|010\rangle+c|001\rangle$ ，其中 $|n_x,n_y,n_z\rangle$ 是表象 $\{H_x,H_y,H_z\}$ 下的基矢。

(3) 分别求 E,l^2,l_z 的期望值;

(4) 写出 t 时刻的量子态。

答案:

(1)

$$\psi_{n_r=0,l=1,m=0}(r,\theta,\varphi)=R_{0,1}(r)Y_{1,0}(\theta,\varphi)=\sqrt{\frac{8}{3\sqrt{\pi}}}\alpha^{5/2}re^{-\alpha^2r^2/2}\sqrt{3/4\pi}\cos\theta$$

$$\psi_{n_r=0,l=1,m=1}(r,\theta,\varphi)=R_{0,1}(r)Y_{1,1}(\theta,\varphi)=\sqrt{\frac{8}{3\sqrt{\pi}}}\alpha^{5/2}re^{-\alpha^2r^2/2}\cdot(-\sqrt{3/8\pi}\sin\theta e^{i\varphi})$$

$$\psi_{n_r=0,l=1,m=-1}(r,\theta,\varphi)=R_{0,1}(r)Y_{1,-1}(\theta,\varphi)=\sqrt{\frac{8}{3\sqrt{\pi}}}\alpha^{5/2}re^{-\alpha^2r^2/2}\sqrt{3/8\pi}\sin\theta e^{-i\varphi}$$

$$\begin{aligned}\psi_{n_x=1,n_y=0,n_z=0}(r,\theta,\varphi)=\psi_1(x)\psi_0(y)\psi_0(z)&=\frac{\sqrt{2\alpha}}{\pi^{1/4}}\alpha x\mathrm{e}^{-\alpha^2x^2/2}\frac{\sqrt{\alpha}}{\pi^{1/4}}\mathrm{e}^{-\alpha^2y^2/2}\frac{\sqrt{\alpha}}{\pi^{1/4}}\mathrm{e}^{-\alpha^2z^2/2}\\&=\frac{\sqrt{2\alpha}}{\pi^{3/4}}\alpha^2r\sin\theta\cos\varphi e^{-\alpha^2r^2/2}\end{aligned}$$

$$\psi_{n_x=0,n_y=1,n_z=0}(r,\theta,\varphi)=\psi_0(x)\psi_1(y)\psi_0(z)=\frac{\sqrt{2\alpha}}{\pi^{3/4}}\alpha^2r\sin\theta\sin\varphi e^{-\alpha^2r^2/2}$$

$$\psi_{n_x=0,n_y=0,n_z=1}(r,\theta,\varphi)=\psi_0(x)\psi_0(y)\psi_1(z)=\frac{\sqrt{2\alpha}}{\pi^{3/4}}\alpha^2r\cos\theta e^{-\alpha^2r^2/2}$$

(2) 在球坐标表象下求此内积，即插入完备性关系:

$$\begin{aligned}&\langle n_r=0,l=1,m=1\mid n_x=0,n_y=1,n_z=0\rangle\\=&\int d^3r\,\langle n_r=0,l=1,m=1\mid r,\theta,\varphi\rangle\,\langle r,\theta,\varphi\mid n_x=0,n_y=1,n_z=0\rangle\\=&\int d^3r\psi^*_{n_r=0,l=1,m=1}(r,\theta,\varphi)\psi_{n_x=0,n_y=1,n_z=0}(r,\theta,\varphi)\\=&\int d^3r\sqrt{\frac{8}{3\sqrt{\pi}}}\alpha^{5/2}re^{-\alpha^2r^2/2}\cdot\left(-\sqrt{3/8\pi}\sin\theta e^{-i\varphi}\right)\frac{\sqrt{2\alpha}}{\pi^{3/4}}\alpha^2r\sin\theta\sin\varphi e^{-\alpha^2r^2/2}\\=&i/\sqrt{2}\end{aligned}$$

(3) 由于基态无简并, 在 (H_x,H_y,H_z) 表象下和 $\left(H,l^2,l_z\right)$ 表象下是同一个态。对第一激发态做从 (H_x,H_y,H_z) 表象到 $\left(H,l^2,l_z\right)$ 表象的表象变换:

$$\begin{pmatrix}-1/\sqrt{2}&\mathrm{i}/\sqrt{2}&0\\1/\sqrt{2}&\mathrm{i}/\sqrt{2}&0\\0&0&1\end{pmatrix}\begin{pmatrix}0\\-i/3\\\sqrt{23}/6\end{pmatrix}=\begin{pmatrix}1/3\sqrt{2}\\1/3\sqrt{2}\\\sqrt{23}/6\end{pmatrix}$$

即 $|\alpha\rangle=1/2|000\rangle+1/3\sqrt{2}|011\rangle+1/3\sqrt{2}|01-1\rangle+\sqrt{23}/6|010\rangle$, 其中 $\mid n_r,l,n$

是表象 $\left(H, \boldsymbol{l}^2, l_z\right)$ 下的基矢, 满足条件:

$$\begin{cases} H\left|n_r lm\right\rangle = \left(2n_r + l + \frac{3}{2}\right)\hbar\omega\left|n_r lm\right\rangle \\ l^2\left|n_r lm\right\rangle = l(l+1)\hbar^2\left|n_r lm\right\rangle \\ l_z\left|n_r lm\right\rangle = m\hbar\left|n_r lm\right\rangle \end{cases}$$

于是

$$\begin{aligned} \langle\alpha|H|\alpha\rangle &= \frac{1}{4}\cdot\frac{3}{2}\hbar\omega + \left(\frac{1}{18}+\frac{1}{18}+\frac{23}{36}\right)\cdot\frac{5}{2}\hbar\omega = \frac{9}{4}\hbar\omega \\ \left\langle\alpha\left|l^2\right|\alpha\right\rangle &= \frac{1}{4}\cdot 0 + \left(\frac{1}{18}+\frac{1}{18}+\frac{23}{36}\right)\cdot 2\hbar^2 = \frac{3}{2}\hbar^2 \\ \left\langle\alpha\left|l_z\right|\alpha\right\rangle &= \left(\frac{1}{4}+\frac{23}{36}\right)\cdot 0 + \frac{1}{18}\cdot\hbar + \frac{1}{18}\cdot(-\hbar) = 0 \end{aligned}$$

(4) t 时刻的量子态 (用表象 (H_x, H_y, H_z) 下的基矢表示):

$$|\alpha(t)\rangle = 1/2e^{-3i\omega t/2}|000\rangle - i/3e^{-5i\omega t/2}|010\rangle + \frac{\sqrt{23}}{6}e^{-5i\omega t/2}|001\rangle$$